

Malanka Wickramasinghe

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Portfolio: malankatharula.github.io/Portfolio
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SUMMARY

I'm a Computer Science undergraduate at the University of Sri Jayewardenepura, and I like being involved in every part of making a product. I design the screens in Figma, build the web and mobile apps myself with React, Next.js, and React Native, and lately most of my own projects have been AI agents and RAG systems built with LangChain, LangGraph, and FastAPI. Along the way I've published a research paper, ranked 1st in my university at IEEEXtreme 18.0, and co-founded the Association of Computing Students.

EDUCATION

University of Sri Jayewardenepura *Jan 2023 - Present*
Bachelor of Computing Honours in Computer Science | Faculty of Computing
Current GPA: 3.66

Ananda College, Colombo, Sri Lanka *Jan 2007 - 2020*
GCE Advanced Level - Physical Science - 3 B's, Z-score: 1.5554
GCE Ordinary Level - 8 A's, 1 C

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, HTML/CSS (Tailwind CSS)	**AI/ML:** LangChain, LangGraph, RAG, AI Agents, Ollama, Groq, ChromaDB, PyTorch, Deep Learning (CNN)
Web Development: React, Next.js, Vite	**Mobile Development:** React Native, Expo, Firebase
Backend & Automation: FastAPI, SQLAlchemy, Alembic, n8n, Docker, Apache Airflow	**UI/UX Design:** Figma, Wireframing, Lo-Fi & Hi-Fi Prototyping
Design & Media: Graphic Design, Video Editing	**Dev Tools:** Git, GitHub, VS Code, Postman, XAMPP

PROJECTS

FacialPrivacyShield - Adversarial Facial Privacy Protection (PyTorch/FastAPI/React) - ongoing - *Final-Year Research* ***link***
Ensemble PGD-based system generating imperceptible perturbations to evade unauthorized face recognition; outperforms Fawkes and LowKey.

USJ Marketplace - UI/UX Design & Mobile App Development (Figma/React Native/Expo/Firebase)
Full-stack e-commerce mobile app for USJ students, taken from UI/UX design through frontend to Firebase backend and database integration.

UI/UX Design Portfolio - FixNow, Salli, Japolic, USJ Marketplace (Figma) ***link***
UI/UX design work for four app projects, from wireframes to hi-fi prototypes; full design documents in the linked Google Drive folder.

ASL-to-Text - Real-Time Sign Language Recognition (Python/CNN) - *Group Project*
Deep learning model for real-time ASL recognition, trained on 200k+ images.

DocuAgent - AI Document Extraction Web App (FastAPI/Groq/Docker) ***link***
AI extracts summaries, key fields, and anomalies from uploaded documents, exporting results to Excel and PDF.

MultiAgentDev - AI Multi-Agent Developer System (LangGraph/Groq/FastAPI) ***link***
Orchestrator routing tasks to specialist Code, Debug, Doc, and Test agents; live demo on Hugging Face Spaces.

CodeRAG - Codebase RAG System (LangChain/ChromaDB/Ollama) ***link***
Q&A over any GitHub repo using code-aware chunking and Llama 3.2.

HR Onboarding Agent - AI Automation Pipeline (n8n/Groq/Gmail OAuth2) ***link***
Single form submission triggers AI-generated welcome emails, task checklists, and automated HR logging.

RESEARCH AND PUBLICATIONS

Published Paper - Scientific Journal of Economics, Management and Sustainability (JEMS), 2024: *"A dual-phase financial strategy for sustainable customer loyalty: Integrating fixed deposits and voucher systems in Sri Lankan supermarket retail."* ***link***

ACHIEVEMENTS & AWARDS

- **IEEEXtreme 18.0** (24-hour global coding competition, 8,000+ teams) - Global Rank: 605, Country Rank: 41, University Rank: 1st (Team CodeDuoPlusOne).
- Only USJ representative invited to IFS Champions Day 2025 at Orion City - exclusive event for Sri Lanka's top-ranked IEEEXtreme coders.
- Co-founder & first Chief Editor, Association of Computing Students (ACS) - USJ; led CodeFuse hackathon, TechXplore industry event, EDUACS seminar series, and ACS eMagazine.

CERTIFICATES

- Introduction to Software Engineering - Udemy (Feb 2024)
- Java Programming Masterclass: Beginner to Master - Udemy (Feb 2024)
- Introduction to Database Management Systems - Udemy (Feb 2024)
- Master Course of Rapid Application Development - Udemy (Feb 2024)
- Introduction to Programming Using Java - Sololearn (Jan 2024)

LEADERSHIP & VOLUNTEERING

Association of Computing Students - University of Sri Jayewardenepura

- **Chief Editor (Apr 2024 - Jun 2025):** Shaped brand identity; led publications, the editorial team, and all major events.

IEEE Computer Society Student Branch - University of Sri Jayewardenepura

- **Design & Marketing Head - J'PURAXTREME 2.0 (2024):** Led all design and marketing efforts for the university's flagship tech event.
- Member - IEEE Student Branch (Oct 2023 - Present).

LANGUAGES

- **Sinhala** (Native)
- **English** (Fluent)

REFERENCES

Mr. Vidura Perera
B.Sc. (Hons) IT, Reading for a PhD,
Lecturer (Asst) - Faculty of Computing,
University of Sri Jayewardenepura.
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